

Here is the partner-facing onboarding document focused strictly on Panasonic Avionics' automation landscape.

Partner Onboarding Guide: Panasonic Avionics Corporation (PAC) – Automation Initiatives

1. Industry and Business Context: Why PAC Cares About Automation

- **The Industry Challenge:** The commercial aviation sector is facing a massive aircraft order backlog, forcing airlines to operate older fleets 1. This has created an unprecedented demand for maintenance, repair, and overhaul (MRO) services, compounded by an industry-wide shortage of certified aviation technicians 2, 3.
- **The PAC Business Challenge:** As the world's leading supplier of in-flight entertainment and communications (IFEC) systems, PAC manages a footprint of over 3,000 aircraft and conducts over 500,000 MRO visits annually 4, 5. To ensure 100% on-time delivery and reduce costly "Aircraft on Ground" (AOG) downtime, PAC must bypass the technician shortage by increasing engineering velocity through proactive automation 5, 6.
- **Overcoming Technical Debt:** Historically, PAC has struggled with a fragmented engineering culture where systems were built without proper controls 7, 8. They are actively standardizing on a Configuration as Code (CaC) and DevOps model to eliminate manual processes and prevent configuration drift 7, 9.

2. Red Hat's Automation Footprint at PAC Red Hat Ansible Automation Platform (AAP) is the core orchestration engine driving PAC's network and infrastructure automation.

- **Current Deployment:** PAC currently runs an on-premise, containerized multi-node AAP environment deployed across three major core sites: New York, Los Angeles, and London 10, 11.
- **Licensing & Growth:** The team is currently operating near the limit of a 100-node license 12, 13. However, due to massive infrastructure ingestion, they have an approved request to expand to an additional 500-node license, creating a significant opportunity to scale our automation efforts with them 12, 14.
- **Custom Execution Environments:** The team actively builds and maintains custom execution environments tailored for specific integrations, including F5, ServiceNow, NetBox, and VMware 10, 15.

3. Key Automation Initiatives and Active Use Cases We are focusing on specific, high-value domains to accelerate PAC's service delivery:

- **Zero-Downtime Event-Driven MRO (ServiceNow Integration):** PAC's internal IT historically relied on a 5-to-10-minute manual polling loop of ServiceNow for change management approvals 5, 10. A massive current initiative is shifting to Event-Driven Ansible (EDA) 16. By utilizing webhooks and the ServiceNow ITSM collection, rulebooks will asynchronously trigger playbooks the exact moment a change request reaches the "implement" state, drastically reducing deployment delays 10, 17.
- **Dynamic Inventory and Single Source of Truth (NetBox):** PAC is establishing NetBox as its Configuration Management Database (CMDB) and single source of truth 10, 18. Ansible is being used to organically manage Configuration Items (CIs) during provisioning and de-provisioning to avoid CMDB duplication 12, 19. A primary goal is

integrating EDA with NetBox webhooks, so events like device additions instantly trigger automated provisioning workflows in AAP 18, 20.

- **F5 Network & DNS Automation:** The team is heavily prioritizing the automation of their global F5 Big IP load balancer infrastructure, which sits at the edge of the network facing the aircraft 18, 21. They are refactoring monolithic legacy playbooks into modular workflows using certified collections to automate nodes, VLANs, route domains, and self-IPs, as well as streamlining their multi-orbit DNS deployments 22-24.
- **Infrastructure as Code (GitLab & Terraform):** To enforce strict DevOps guardrails, PAC uses GitLab as their source of truth for configuration templates and variables 25, 26. Ansible integrates with their Terraform workflows to establish a reliable, repeatable deployment model across their environments 9, 27.

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